Absolute value equations

1 Write an absolute value equation for each of the following:

a Representing all numbers x whose distance from 0 is 9 units.

b $5x - 2 = 3$ or $5x - 2 = -3$

2 Determine how many solutions there are for each of the following absolute value equations:

a $|x| = a$, where a is a positive number.

b $|x| = -8$

3 For what value(s) of b does the equation $|x| = b$ have only one solution?

4 Roxanne performed the following steps to solve the equation $4|x| = -2$:

1. $4|x| = -2$
2. $4x = \pm 2$
3. $x = \pm \dfrac{1}{2}$

Which, if any, of the numbers that Roxanne wrote down as solutions actually satisfy the equation $4|x| = -2$? Explain your answer.

5 Solve the following equations:

a $|x| = 8$

b $|x| = 10$

c $9|x| = 4$

d $2|x| = 10$

e $\dfrac{|x|}{2} = 20$

f $\dfrac{|x|}{7} = 6$

g $\dfrac{2|x|}{3} = 4$

h $-\dfrac{5|x|}{2} = -1$

6 Solve the following equations:

a $|x + 3| = 7$

b $|x + 5| = 6$

c $|x - 3| = 11$

d $|1 - x| = 3$

e $|7 - x| = 15$

f $|4 - x| = 10$

7 Solve the following equations:

a $|2x+6|=10$

b $|3x+6|=9$

c $-3\,|x+5|=-9$

d $|2x+9|=11$

e $|3x-7|=8$

f $|5x-1|=9$

g $|6x+10|=14$

h $|8x+15|=7$

8 Solve the following equations:

a $|2x|-5=4$

b $|3x|+8=29$

c $7-|4x|=5$

d $|4x-8|+1=13$

e $|-3.9x|+3=22.5$

f $|2x+9|=x+6$